

Problem Set 1

PCC190 - Secure Machine Learning

1. Why is machine learning important nowadays? What is its concomitant risk?
2. How does the book define the problem of secure machine learning? Are there other possible definitions? What are they?
3. What are the key strengths of learning approaches, and how can they be subverted?
4. What is the main reason machine learning techniques are being applied to a growing number of systems?
5. What is a malfeasance detection problem?
6. Give examples of malfeasance detection problems that can be solved with machine learning techniques, and explain how those techniques solve them.
7. According to the book, what is the primary vulnerability in learners that attackers can exploit?
8. What are the two fundamental questions regarding machine learning security?
9. Describe how a spam filter based on machine learning works.
10. Describe the operation of a network anomaly detector based on PCA. How can an attacker evade such a system?
11. Why would privacy be a concern in machine learning systems? Give examples.
12. What are the steps involved in the evaluation of a computer system's security?
13. What is computer security concerned about?
14. In your own words, describe the following classical security principles:
 - a) Proactive analysis
 - b) Kerckhoffs's Principle
 - c) Conservative design
 - d) Threat modeling (how is a threat model constructed?)